

Ankur Desai

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EDUCATION

Michigan State University

B.S. in Computer Science

Expected Graduation, 2027

East Lansing, MI

- **GPA:** 3.5 / 4.0
- **Relevant Coursework:** Linear Algebra, Discrete Mathematics, Object-Oriented Software Design, Computer Organization and Architecture, Data Structures & Algorithms

EXPERIENCE

MindMosaic

Software Engineer

Aug. 2025 – Nov. 2025

East Lansing, MI

- Partnered with team on requirements and introduced Neo4j to offload recursive queries from relational database (PostgreSQL), cutting p95 latency 48% (850ms → 440ms) and eliminating SQL join overhead.
- Optimized database queries via Cypher relationship-type indexing; traversals ran 3× faster than SQL self-joins, removing full-table scans.
- Integrated OAuth 2.0 (Google + GitHub) with httpOnly JWT cookies, refresh rotation, and user-scoped graph access control.
- Implemented GitLab CI/CD spanning full SDLC and Agile sprints: build, test, and containerized deployment with branch-protected merges, enforcing 85% coverage across 35+ tests.

Claude Builder Club @ MSU

Vice President

Oct. 2025 – Present

East Lansing, MI

- Eliminated manual onboarding by automating member provisioning (GitHub API + Slack SDK) for cross-functional teams, granting repo and workspace access in 30 seconds and cutting setup time 97% across 100+ members.
- Replaced paper sign-in sheets with a QR-driven check-in system backed by AWS S3 token caching, enabling real-time attendance tracking and cutting processing time by 80%.
- Enforced production-grade quality via GitHub Actions CI/CD with 90% coverage gates across 20+ tests.

PROJECTS

Mira — Agentic Voice Assistant | *Next.js, FastAPI, MongoDB, Redis, Celery* May 2025 – Nov. 2025

- Engineered agentic voice assistant (TypeScript, Next.js) that resolves implicit speaker intent and dispatches real tasks across concurrent voice streams.
- Reduced REST API response time by 73% by offloading RAG pipeline to async Redis/Celery task queue.
- Built Electron + Next.js client with embedded HTTP server for OAuth callbacks; IPC and webhook server deliver sub-50ms event propagation.
- Deployed FastAPI on AWS Lambda with lazy-loaded ML models (Whisper, Sentence-BERT, spaCy), reducing cold-start 86% (6,643ms → 930ms).

WizViz — SpartaHack X | *Python, MediaPipe, OpenCV, NumPy, TensorFlow*

Feb. 2025

- Won Interactive Media Track: Gesture-controlled wizard duel where full-body movements cast spells, block attacks, and trigger abilities against an AI opponent.
- Built full-body gesture pipeline using MediaPipe Pose to map 33 skeletal landmarks into game actions; optimized inference loop to sustain 60 FPS with sub-20ms input lag.

TECHNICAL SKILLS

Languages Python, Java, JavaScript/TypeScript, C#, SQL, Cypher, Golang, C++, Bash
Frameworks Spring Boot, FastAPI, React, Next.js, Node.js, Redis, Agile, REST APIs
Databases PostgreSQL, MySQL, Oracle, Neo4j, MongoDB
Cloud & Infrastructure AWS (EC2, RDS, S3), Docker, GitLab CI/CD, CI/CD Pipelines